RES_UNIT-I_SOLAR ENERGY

Introduction

Renewable energy is energy produced from sources that do not deplete or can be replenished or refilled within a human's life time. The most common examples of renewable energy sources include wind, solar, geothermal, biomass, and hydropower. Non-renewable energy comes from sources that will run out or will not be replenished in our lifetime or even in many lifetimes. Most of the non-renewable energy sources are fossil fuels, which influence the environment greatly and contribute to harmful global warming and climate change. Renewable energy is sustainable as it originates from sources that are inexhaustible (unlike fossil fuels). Despite of many advantages renewable energy sources have certain limitations like higher capital cost, intermittency, storage capabilities, geographic limitations, etc., which make them inevitable.

Environmental consequences of fossil fuel use

Fossil fuels are formed from the fossilized, buried remains of plants and animals that lived millions of years ago so they are named accordingly. Fossil fuels, which include coal, natural gas, petroleum, shale oil, and bitumen, are the main sources of heat and electrical energy. All these fuels contain the major constituents like carbon, hydrogen, oxygen and other materials like metal, sulphur and nitrogen compounds. During the combustion process different pollutants like fly ash, sulphur oxides (SO_2 and SO_3), nitrogen oxides ($NO_x = NO_2 + NO$) and volatile organic compounds are emitted. Gross emission of these pollutants constitutes to atmospheric pollution and can affect human beings and environment.

Atmospheric Pollution

Atmospheric pollution occurs in many forms but can generally be thought of as gaseous and particulate contaminants that are present in the earth's atmosphere. Chemicals discharged into the air that have a direct impact on the environment are called primary pollutants. These primary pollutants sometimes react with other chemicals in the air to produce secondary pollutants. The most commonly found air pollutants are oxides of Sulphur, oxides of nitrogen, oxides of carbon, hydrocarbons, particulates (fly ash).

Oxides of Sulphur (SO2)

Sulphur dioxide (SO_2) is a colourless gas with a sharp, irritating odour. It is produced by burning fossil fuels and by the smelting of mineral ores that contain sulphur. Erupting volcanoes can be a significant natural source of sulphur dioxide emissions.

Environmental effects

When sulphur dioxide combines with water and air, it forms sulphuric acid, which is the main component of acid rain. Acid rain can:

- Cause deforestation
- Acidify waterways to the detriment of aquatic life
- Corrode building materials and paints.

Health effects

- Sulphur dioxide affects the respiratory system, particularly lung function and can irritate the eyes.
- Sulphur dioxide irritates the respiratory tract and increases the risk of tract infections.
- It causes coughing, mucus secretion and aggravates conditions such as asthma and chronic bronchitis.

Oxides of Nitrogen (NOx)

The term nitrogen oxides (NOx) describes a mixture of nitric oxide (NO) and nitrogen dioxide (NO_2), which are gases produced from natural sources, motor vehicles and other fuel burning processes. Nitric oxide is colourless and is oxidised in the atmosphere to form nitrogen dioxide. Nitrogen dioxide has an odour and is an acidic and highly corrosive gas that can affect our health and environment. In poorly ventilated situations, indoor domestic appliances such as gas stoves and gas or wood heaters can be significant sources of nitrogen oxides.

Environmental and health effects of nitrogen oxides

- Elevated levels of nitrogen dioxide can cause damage to the human respiratory tract and increase a person's vulnerability to respiratory infections and asthma.
- Long-term exposure to high levels of nitrogen dioxide can cause chronic lung disease.
- It may also affect the senses of smell an odour.
- High levels of nitrogen dioxide are also harmful to vegetation, damaging foliage, decreasing growth or reducing crop yields.
- Nitrogen dioxide can fade and discolour furnishings and fabrics, reduce visibility and react with surfaces.

Oxides of Carbon (CO, CO2)

Carbon monoxide is a colourless, odourless gas formed when substances containing carbon (such as petrol, gas, coal and wood) are burned with an insufficient supply of air. Motor vehicles are the main source of carbon monoxide pollution in urban areas.

Health effects

- Carbon monoxide has serious health impacts on humans and animals.
- When inhaled, the carbon monoxide bonds to the haemoglobin in the blood in place of oxygen to become carboxyhaemoglobin. This reduces the oxygen-carrying capacity of the red blood cells and decreases the supply of oxygen to tissues and organs, especially the heart and brain.
- For people with cardiovascular disease, this can be a serious problem.
- The effects are reversible, so symptoms decrease gradually when exposure to carbon monoxide stops.

Hydrocarbons

A hydrocarbon is any compound that consists of carbon and hydrogen atoms. They are organic compounds. Because of the unique covalent nature of carbon, there are thousands upon thousands of hydrocarbons in the world. Gasoline, petroleum, coal, kerosene, charcoal, natural gas, etc., are all a form of hydrocarbons.

Environmental and health effects of hydrocarbons

- These substances contribute to the greenhouse effect and climate change
- Deplete the ozone
- Reduce photosynthetic ability of plants
- Increase occurrences of cancer and respiratory disorders in humans.

India uses about 500 million T of coal every year to produce electricity, about 3.6 trillion cubic feet of natural gas for power, chemicals and fertilizers and over 160 million T of oil for transport and industry.

Particulates (Fly Ash)

Fly ash is composed of tiny, airborne particles and is thus considered as a type of particulate matter or particle pollution. Fly ash contains different trace elements (heavy metals).

Environmental and health effects of fly ash

Wet ash ponds can pollute groundwater and if ingested, the arsenic contaminated water increases a person's risk of developing cancer.

Inhalation or ingestion of the toxins in fly ash can have impacts on the nervous system, causing cognitive defects, developmental delays, and behavioural problems while also increasing a person's chance of developing lung disease, kidney disease, and gastrointestinal illness.

When ash is disposed in dry landfills or wet ponds, there are associated environmental effects. Wet surface impoundments account for a fifth of coal ash disposal. These wet impoundments can be an issue if they do not have proper liners for the landfill or pond to prevent leaking and leaching. Both leaking and leaching lead to groundwater contamination.

Leaching is a process that occurs when fly ash is wet, and it simply means that the toxic components of the ash dissolve out and percolate through water. This groundwater contamination can be harmful to human health if the groundwater is a source of drinking water. In addition to leaching, fly ash toxics are able to travel through the environment as a result of erosion, runoff, or through the air as fine dust. The fact that the chemicals in the ash can escape and move through the environment is what makes fly ash harmful.

Green House Gas Emissions from Various Energy Sources

Greenhouse gases are gases in earth's atmosphere that trap heat. They let sunlight pass through the atmosphere, but they prevent the heat that the sunlight brings from leaving the atmosphere.

Most of the emissions of human-caused (anthropogenic) greenhouse gases come primarily from burning fossil fuels like coal, hydrocarbon gas liquids, natural gas and petroleum, for energy use. Global warming or climate change has been observed for around 150 years and is a growth in this phenomenon.

The other GHG that are emitted as a result of human activity are

- Methane (CH_4), which comes from landfills, coal mines, agriculture, and oil and natural gas operations
- Nitrous oxide (N_2O), which comes from using nitrogen fertilizers and certain industrial and waste management processes and burning fossil fuels
- High global warming potential (GWP) gases, which are human-made industrial gases
- Hydrofluorocarbons (HFCs)
- Perfluorocarbons (PFCs)

- Sulfur hexafluoride (SF_6)
- Nitrogen trifluoride (NF_3)

Importance of renewable sources of energy

Renewable energies are obtained from sources of clean, inexhaustible and increasingly competitive energy. They differ from fossil fuels principally in their diversity, abundance and potential for use anywhere on the planet. In addition, they produce neither greenhouse gases – which cause climate change – nor polluting emissions. Their costs are also falling at a sustainable rate, whereas the general cost trend for fossil fuels is in the opposite direction in spite of their present volatility.

Renewable energies received assistance from the international community through the Paris Accord signed at the World Climate Summit held in Paris on December 2015. The agreement, which will enter into force in 2020, establishes, for the first time in history, a binding global objective. Nearly 200 signatory countries pledged to reduce their emissions so that the average temperature of the planet at the end of the current century remains well below 2° C, the limit above which climate change will have more catastrophic effects. However, the aim is to keep it to 1.5° C. Other importance of renewable energy are:

Indispensable partner in the fight against climate change: Renewables do not emit greenhouse gases in energy generation processes, making them the cleanest, most viable solution to prevent environmental degradation. Most renewable energy sources produce little to no global warming emissions. Even when including "life cycle" emissions of clean energy (i.e, the emissions from each stage of manufacturing, installation, operation, decommissioning), the global warming emissions associated with renewable energy are minimal.

Inexhaustible: Compared to conventional energy sources such as coal, gas, oil and nuclear - reserves of which are finite - clean energies are just as available as the sun from which they originate and adapt to natural cycles, hence their name "renewables". This makes them an essential element in a sustainable energy system that allows development today without risking that of future generations.

Reducing energy dependence: The indigenous nature of clean sources gives local economies an advantage and brings meaning to the term "energy independence". Dependence on fossil fuel imports results in subordination to the economic and political short-term goals of the supplier country, which can compromise the security of energy supply.

Increasingly competitive: The main renewable technologies – such as wind and solar photovoltaic – are drastically reducing their costs, such that they are fully competitive with conventional sources in a growing number of locations. Economies of scale and innovation are already resulting in renewable energies becoming the most sustainable solution, not only environmentally but also economically, for powering the world. Renewable energy is providing affordable electricity across the country right now, and can help stabilize energy prices in the future.

Benefiting from a favourable political horizon: The international community has understood its obligation to firm up the transition towards a low-carbon economy in order to guarantee a sustainable future for the planet. International consensus in favour of the "de-carbonization" of the economy constitutes a very favourable framework for the promotion of clean energy technologies.

Improved public health: Wind, solar, and hydroelectric systems generate electricity with no associated air pollution emissions. Geothermal and biomass systems emit some air pollutants, though total air emissions are generally much lower than those of coal- and natural gas-fired power plants. In addition, wind and solar energy require essentially no water to operate and thus do not pollute water resources or strain supplies by competing with agriculture, drinking water, or other important water needs.

Jobs and other economic benefits: Compared with fossil fuel technologies, which are typically mechanized and capital intensive, the renewable energy industry is more labour intensive. Solar panels need humans to install them; wind farms need technicians for maintenance. This means that, on average, more jobs are created for each unit of electricity generated from renewable sources than from fossil fuels.

Reliability and resilience: Wind and solar are less prone to large-scale failure because they are distributed and modular. Distributed systems are spread out over a large geographical area, so a severe weather event in one location will not cut off power to an entire region. Modular systems are composed of numerous individual wind turbines or solar arrays. Even if some of the equipment in the system is damaged, the rest can typically continue to operate. Wind and solar photovoltaic systems do not require water to generate electricity and can operate reliably in conditions that may otherwise require closing a fossil fuel-powered plant due to water scarcity.

Sl. No	Renewable Energy	Advantages
1	Solar energy (From the sun)	• Sunlight does not produce any wastes or pollutants for environment.

		• It is free to collect sunlight as it is always present
2	The Wind	• The wind does not produce any wastes or pollutants for environment. • It takes up little ground space
3	Hydropower	• Hydropower is considered as inexpensive source. • It does not leave any harmful chemicals as waste.
4	Biomass	• Growing biomass crops use up carbon dioxide and increase • Biomass is always available, thus, it can be used as renewable resource.
5	Geothermal Energy	• For heating and cooling, geothermal heat pump systems use 25% to 50% less electricity than conventional systems. • Biomass is always available and can be used as a renewable resource

Types of RE sources

Alternative or renewable energy comes from natural processes that can reliably produce cheap energy with minimal impact to the environment. The most popular renewable energy sources currently are:

- Solar energy
  Energy obtained from sunlight using solar panels or thermal collectors. Used for electricity generation, water heating, and drying.
- Wind energy
  Energy produced by converting wind motion into electrical power using wind turbines.
- Hydro energy
  Energy generated from flowing or falling water using turbines in dams or rivers.
- Tidal energy
  Energy produced from ocean tides.
- Geothermal energy
  Energy obtained from heat stored beneath the Earth's surface.
- Biomass energy
  Energy derived from organic materials such as wood, agricultural waste, and biogas.
- Hydrogen
  Hydrogen fuel cells convert the potential chemical energy of hydrogen into electricity.

Introduction to Solar Energy:

A solar cell or photovoltaic cell is an electrical device that converts the energy of light directly into electricity by photovoltaic effect.

Photovoltaic conversion is the process of direct conversion of light into electricity at the atomic level.

Some materials exhibit a property known as the photoelectric effect that causes them to absorb photons of light and release electrons. When free electrons are captured, electricity is generated

They are made of special materials that are semiconductors.

These semiconductors produce electricity when sun light falls onto its surface.

It is 5 to 11 times more expensive to produce electricity from the sun than it is from fossil and nuclear fuels.

The cost of the solar cell technology that uses expensive semiconductor materials to convert sunlight directly into DC electricity. Moreover, the solar cell efficiency is not very encouraging. Practical efficiency of solar cells lies somewhere in between 10% and 25%.

Need of a Solar Cell: The development of solar cell use has been advocated because of the following needs

1. For low maintenance, long lasting sources of electricity suitable for places remote from both the main electricity grid and the people, that is, satellites, remote site water pumping, out back telecommunications stations, and lighthouses;
2. For cost effective power supplies for people living in the remote areas that have no access to the main electricity grid; for example, aboriginal settlements, outback sheep and cattle stations, and some home sites in grid-connected areas.
3. For non-polluting and silent sources of electricity, i.e., tourist sites, caravans, and campers
4. For a convenient and flexible source of small amounts of power, that is, calculators, watches, light meters, and cameras;

5. For renewable and sustainable power, as a means of reducing global warming.

Solar Photovoltaic systems: Basic Principle of SPV conversion

A photovoltaic (PV) cell is an energy harvesting technology that converts solar energy into useful electricity (DC) through a process called the photovoltaic effect. It is made up of semiconductor materials such as silicon, gallium arsenide and cadmium telluride, etc. These cells vary in size ranging from about 0.5 inches to 4 inches. There are different types of PV cells which all use semiconductors to interact with incoming photons from the Sun in order to generate an electric current.

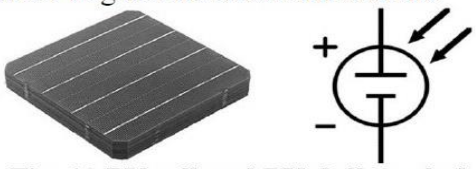


Fig. 19 PV cell and PV Cell symbol

Photovoltaic cell concepts: Cell
Layers of a PV Cell

A photovoltaic cell is comprised of many layers of materials, each with a specific purpose. The most important layer of a photovoltaic cell is the specially treated semiconductor layer. It is comprised of two distinct layers of p-type and n-type, and is what actually converts the Sun's energy into useful electricity through a process called the photovoltaic effect. On either side of the semiconductor is a layer of conducting material which "collects" the electricity produced. Note that the backside or shaded side of the cell can afford to be completely covered in the conductor, whereas the front or illuminated side must use the conductors sparingly to avoid blocking too much of the Sun's radiation from reaching the semiconductor.

The final layer which is applied only to the illuminated side of the cell is the anti-reflection coating. Since all semiconductors are naturally reflective, reflection loss can be significant. The solution is to use one or several layers of an anti-reflection coating (similar to those used for eyeglasses and cameras) to reduce the amount of solar radiation that is reflected off the surface of the cell.

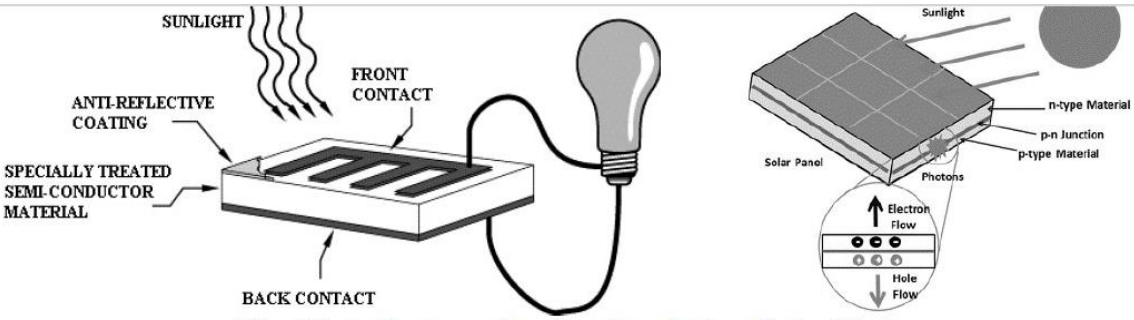


Fig. 20. A diagram showing the photovoltaic effect.

Photovoltaic Effect

The photovoltaic effect is a process that generates voltage or electric current in a photovoltaic cell when it is exposed to sunlight. These solar cells are composed of two different types of semiconductors—a p-type and an n-type that are joined together to create a p-n junction. By joining these two types of semiconductors, an electric field is formed in the region of the junction as electrons move to the positive p-side and holes move to the negative n-side. This field causes negatively charged particles to move in one direction and positively charged particles in the other direction.

Light is composed of photons, which are simply small bundles of electromagnetic radiation or energy. When light of a suitable wavelength is incident on these cells, energy from the photon is transferred to an electron of the semiconducting material, causing it to jump to a higher energy state known as the conduction band. In their excited state in the conduction band, these electrons are free to move through the material, and it is this motion of the electron that creates an electric current in the cell.

- **Cell:** A photovoltaic cell is the most basic unit of a solar PV system - solar cells can be either monocrystalline or polycrystalline, and their key characteristic is that they produce a voltage output when exposed to light. It is important to note that although they are normally called "solar cells", they can respond to any type of light. Each cell produces approximately 1/2 a volt and a solar module can have any number of solar cells.

- **Module:** A photovoltaic module consists of multiple PV cells connected in series to provide a higher voltage output. PV modules are manufactured in standard sizes. A solar module designed for charging a 12 volt battery will typically have 36 solar cells while the typical residential grid connected system uses solar modules with 60 solar cells. For large commercial and utility scale solar systems, solar modules will have typically 72 solar cells. By increasing the number of solar cells the module voltage and wattage increases. The term solar panel is sometimes used interchangeably with solar module. The main difference is that some solar panels models are composed of multiple modules mounted together.
- **Array:** A photovoltaic array is a system composed of multiple PV modules. They can be connected in one or more series circuits, which are connected to a combiner box to provide a single direct-current output. This output can be used to charge batteries, power DC loads, or fed to an inverter to provide an AC voltage for home appliances or exporting to the electric grid.

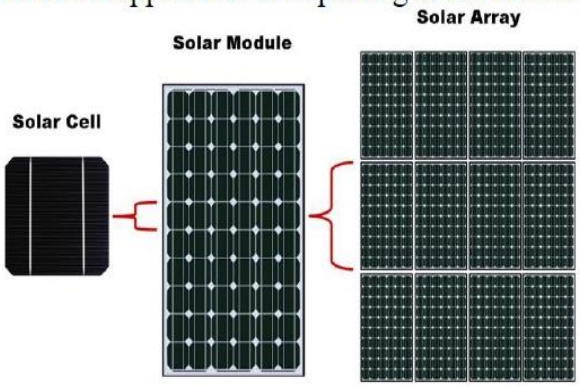


Fig.21. Cell module and array

Solar Energy:

- The energy radiation from the sun is electro-megnetic waves reaching the planet earth in three spectral reagions:

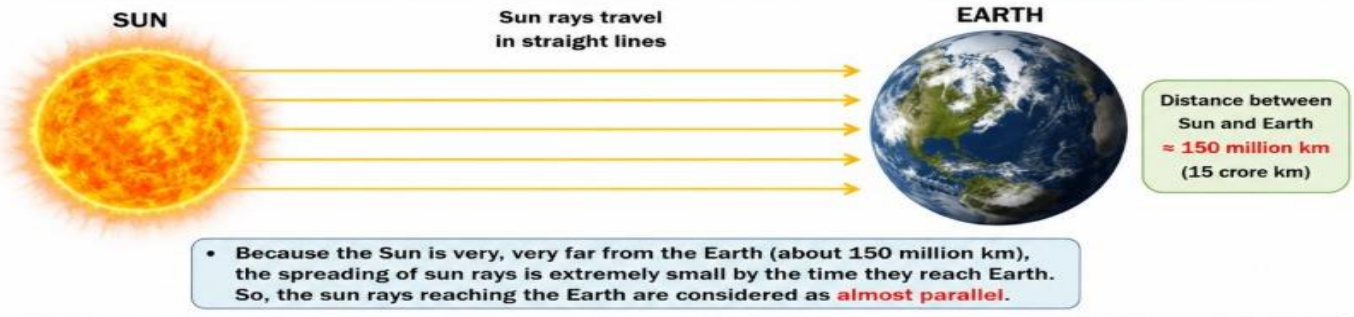
i) Ultraviolet: 6.4% ($\lambda \leq 0.38$ μm)

ii) Visible: 48 % (0.38 μm $\leq \lambda \leq 0.78$ μm)

iii) Infrared: 45.6 % ($\lambda \geq 0.78$ μm) of total energy

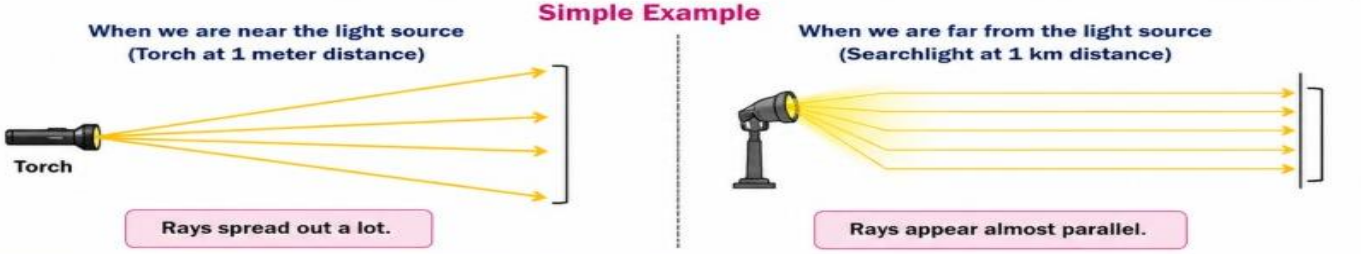
- Due to large diatance between the sun and the earth the bean of radiation received from the sun on the earth is almost parallel.

Due to the large distance between the Sun and the Earth, the beam of radiation received from the Sun on the Earth is almost parallel.



- Because the Sun is very, very far from the Earth (about 150 million km), the spreading of sun rays is extremely small by the time they reach Earth. So, the sun rays reaching the Earth are considered as almost parallel.

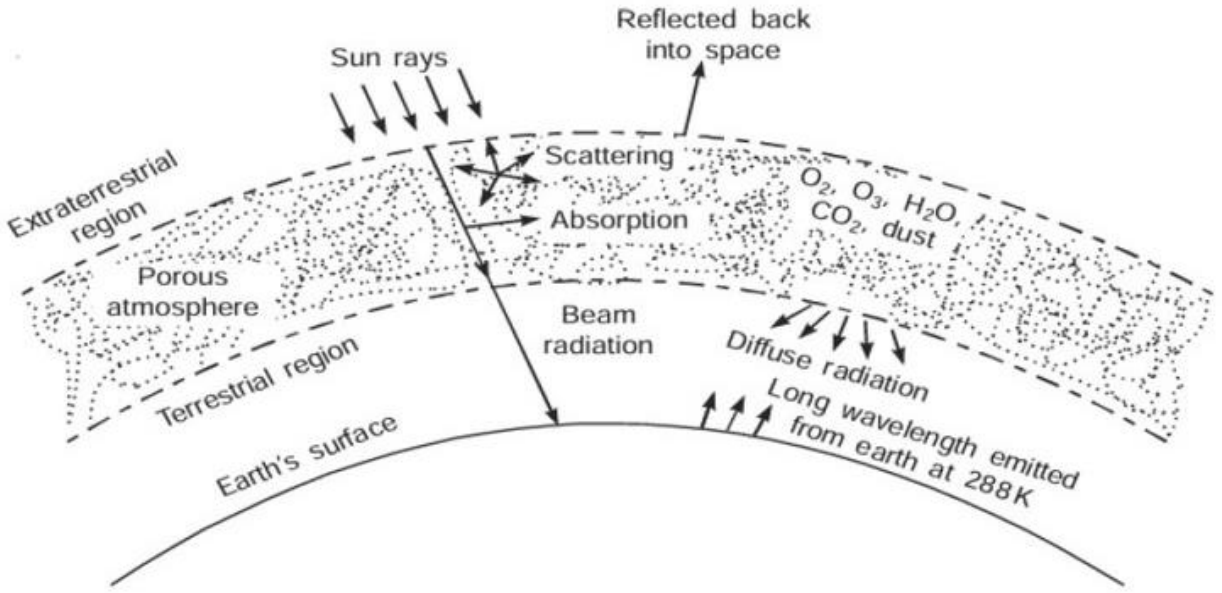
Simple Example



Conclusion

Since the Sun is extremely far from the Earth, the sun rays that reach Earth can be assumed as straight and parallel.





Solar Radiation is the electromagnetic energy emitted by the Sun and received by the Earth.

- **Direct radiation:** Solar radiation that reaches to the surface of the earth with out being diffused is called direct beam radiation.
- **Diffused radiation:** As sunlight passes through the atmosphere, some of it is abosrbed, scattered, and reflected by the air molecules, water vapour, cloud, dust and pollutants from the power plants, forest fires etc. This is called diffusion radiation.

Solar Radiation:

- **Total radiation:** The sum of beam and diffuse radiations (Ib + Id) is referred to as total radiation. When measured at a location on the earth's surface, it is called solar insolation at the place. When measured on a horizontal surface, it is called global radiation.
- **Sun at zenith:** It is the position of the sun directly overhead.

Not all sunlight reaches the Earth's surface.
Some energy is:

- Reflected back into space
- Absorbed by gases
- Scattered by dust and clouds

Only the remaining energy reaches the ground.

Direct Radiation (Beam Radiation)
Solar rays that reach the Earth without changing direction.
Example:
Sunny day with a clear sky.
Characteristics
✓ Highest intensity
✓ Travels in a straight line
✓ Produces sharp shadows

Diffuse Radiation
Solar rays scattered by clouds, dust, and air molecules before reaching Earth.
Example:
Cloudy day.
Characteristics
✓ Comes from all directions
✓ Lower intensity
✓ No sharp shadows

Reflected Radiation
Radiation reflected by buildings, soil, water, snow, etc.
Example:
Sunlight reflected from a white wall.

Global Solar Radiation = Direct + Diffuse Radiation

Atmosphere Acts Like a Filter
Atmosphere contains:

- Oxygen
- Nitrogen
- Carbon dioxide
- Dust
- Water vapour
- Clouds

These reduce solar radiation

Absorption
Some gases absorb sunlight.
Examples:

- Ozone absorbs UV rays.
- Water vapour absorbs infrared rays.

Scattering

- Dust particles scatter sunlight.

Factors Affecting Solar Radiation

1. Latitude
Closer to the Equator: More sunlight
Closer to the Poles: Less sunlight

2. Time of Day
Morning-Low radiation
Noon-Maximum radiation
Evening-Low radiation

3. Season
Summer-More radiation
Winter-Less radiation

4. Clouds
More clouds-Less solar radiation

5. Dust Pollution
Dust blocks sunlight.

6. Altitude
Higher places receive more solar radiation because the atmosphere is thinner.

Advantages of High Solar Radiation

More electricity
Better crop drying
Better water heating
Higher PV efficiency

Disadvantages

Depends on
Weather
Clouds
Dust
Night time

Radiation at Earth's Surface

Average radiation reaching the Earth's surface
Approximately 1000 W/m² on a clear sunny day around noon.

Solar Constant : The sun, being at a very large distance from the earth, solar rays subtend an angle of only 32 minutes on earth.

- The solar constant, Isc, is the energy from the sun received on a unit area perpendicular to the solar rays at the mean distance from the sun outside the atmosphere.

$$I_n = I_{sc}\left(1 + 0.033\cos\frac{360n}{365}\right)$$

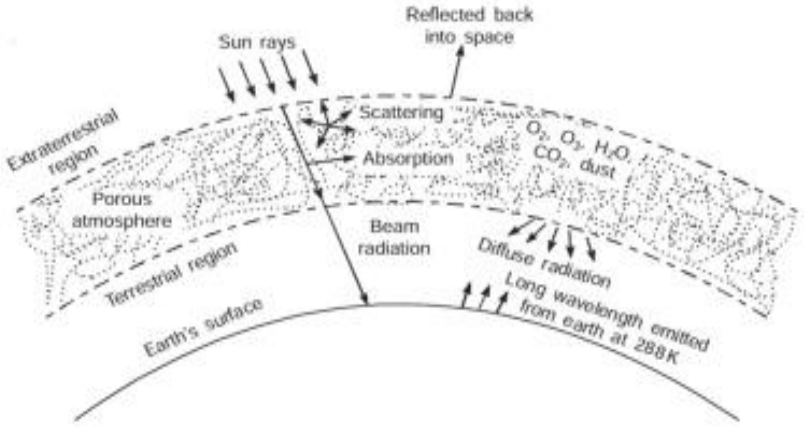
- Based on the experimental measurements, the standard value of the solar constant is 1367 W/m^2 .
- In terms of other units, Isc = 432 Btu//h or 4.921 MJ/ m^2 /h.

Solar Constant is the solar energy received on a unit area outside the Earth's atmosphere when the Sun's rays are perpendicular.

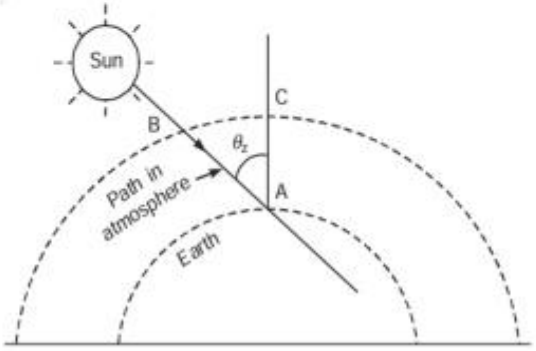
This value is measured outside the atmosphere.
Because of atmospheric losses, the Earth surface receives less.

Solar Radiation:

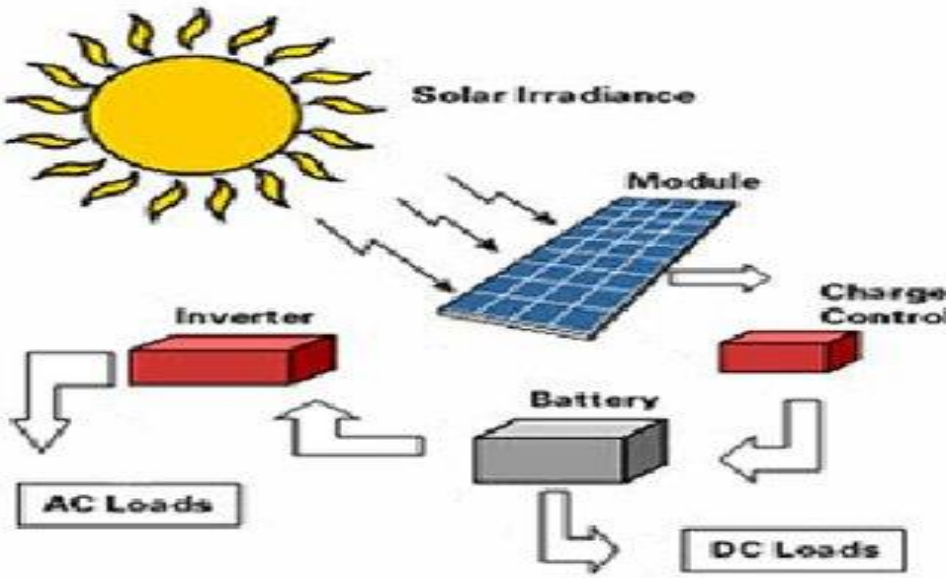
- Solar radiations pass through the earth's atmosphere and are subjected to scattering and atmospheric absorption.
- Short wave ultraviolet rays are absorbed by ozone and long wave infrared rays are absorbed by CO2 and water vapours.
- Scattering is due to air molecules, dust particles and water droplets that cause attenuation of radiation as detailed



- **Irradiance (W/m2):** The rate of incident energy per unit area of a surface is termed irradiance.
- **Albedo:** The earth reflects back nearly 30% of the total solar radiant energy to the space by reflection from clouds, by scattering and by reflection at the earth's surface. This is called the albedo of the earth's atmosphere system.



Solar energy



Sunlight is a renewable resource, and its most direct use is achieved by capturing the sun's energy. A variety of solar energy technologies are used to convert the sun's energy and light into heat: illumination, hot water, electricity and (paradoxically) cooling systems for businesses and industry.
Photovoltaic (PV) systems use solar cells to convert sunlight into electricity. Solar hot water systems can be used to heat buildings by circulating water through flat-plate solar collectors. Mirrored dishes that are focused to boil water in a conventional steam generator can produce electricity by concentrating the sun's heat. Commercial and industrial buildings can also leverage the sun's energy for larger-scale needs such as ventilation, heating, and cooling. Finally, thoughtful architectural designs can passively take advantage of the sun as a source of light for heating and cooling. Home owners, businesses and government entities can take advantage of the benefits of solar power in many ways: Install a home solar system or commercial solar panels; construct or retrofit a building to incorporate solar hot water, cooling or ventilation systems; design from scratch structures that take advantage of the sun's natural attributes for passive heating and lighting.

The basic principle behind both solar panel – solar photovoltaic (PV) and solar thermal – is the same. They absorb raw energy from the sun and use it to create usable energy. In solar PV systems this is through the creation of electricity, whereas thermal systems are used directly for heating water or air. The amount of solar radiation on the earth surface can be instrumentally measured using Pyrheliometer, Pyranometer, Photoelectric sunshine recorder and many instruments. Solar thermal power plants collect and concentrate sunlight to produce the high temperature heat needed to generate electricity. Thermal energy storage (TES) is a technology that stocks thermal energy by heating or cooling a storage medium so that the stored energy can be used at a later time for heating and cooling applications and power generation. A photovoltaic module consists of multiple PV cells connected in series to provide a higher voltage output. A photovoltaic array is a system composed of multiple PV modules. They can be connected in one or more series circuits, which are connected to a combiner box to provide a single direct-current output.

Solar Radiation

Solar radiation, often called the solar resource, is a general term for the electromagnetic radiation emitted by the sun. The sun emits electromagnetic radiations as a black body having a surface temperature of about 6000 K. This is because of the nuclear reaction running in it, where the sun is converting hydrogen into helium. The radius of the sun amounts to 1.39×10^9 m. The total radiation power received from the sun on a unit area perpendicular to the sun rays at the mean earth sun distance, termed an astronomical unit, is called the solar constant (SC), where 1 astronomical unit = 1AU = 1.496×10^{11}m.

Operation of PV Systems

Solar cells produce direct current (DC), therefore they are only used for DC equipments. If alternating current (AC) is needed for AC equipment or backup energy is needed, solar photovoltaic systems require other components in addition to solar modules. These components are specially designed to integrate into solar PV system. The components of solar photovoltaic system are shown in the figure.

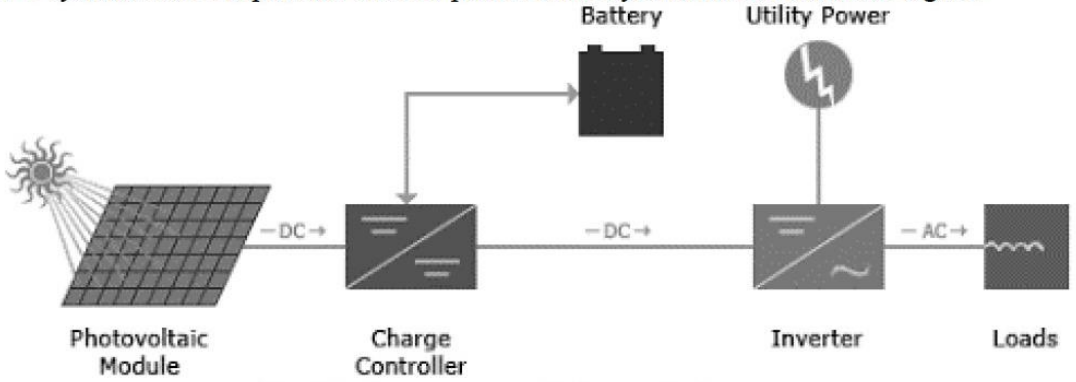


Fig. 22. Components of Photovoltaic system

1. **Solar or Photovoltaic Module** is the essential component of any solar PV system that converts sunlight directly into DC electricity.
2. **Solar Charge Controller** regulates voltage and current from solar arrays, charges the battery, prevents battery from overcharging and also performs controlled over discharges.
3. **Battery** stores the energy produced from solar arrays for using when sunlight is not visible, night time or other purposes.
4. **Inverter** is a critical component of any solar PV system that converts DC power output of solar arrays into AC for AC appliances.
5. **Lightning protection** prevents electrical equipment from damages caused by lightning or induction of high voltage surge. It is required for the large size and critical solar PV systems, which include efficient grounding.

Net metering is a billing mechanism that credits solar energy system owners for the electricity they add to the grid.

Limitations of solar

1) **Higher Costs than Fossil Energy Forms** –It has been estimated that solar power costs fall by 20% for every 100% increase in supply. The Solar Cost Curve has declined massively in the last 2 years as cheap Chinese solar production has made solar panel costs come down by 50%. Note in the next 4-5 years expect an average decline of around 10% per year which would make solar energy competitive with fossil fuel energy in most parts of the world. Current solar power costs between 15-30/Kwh depending on the solar radiation of the particular location, type of technology used etc.

2) **Intermittent Nature** – One of the biggest problems of Solar Power is that it is intermittent in nature as it generates energy only when the sun shines. This problem can be solved with energy storage however this leads to additional costs. Smart Grids and Cheaper Energy Storage in the future should allow even higher penetrations of Wind and Solar Power possible.

3) **High Capital Investment** – A Solar Plant can cost around 450 lakhs to be spent in building 1 Megawatt. This is said to be too high, however the costs of energy can only be compared by Levelized Cost of Energy (LCOE) which calculates the cost of energy over the lifetime calculating the capex, fuel costs, maintenance, security and insurance costs. While it is true that the initial capital investment for solar power is quite high, the lifecycle cost of solar energy is not that high.

4) **Cannot be Built Anywhere** – This disadvantage of Solar Energy is present with other forms of Energy as well. Some forms of Energy are just better suited to some places. For example you can't build a nuclear plant on top of an earthquake prone region, you can't build a wind farm near the Dead Sea., etc.,

Limitations of Solar Energy

Depends on sunlight (weather dependent)
No power generation at night
Seasonal variations affect output
Energy storage is necessary

Renewable energy sources are essential for sustainable development, and among them, solar energy is the most promising due to its availability, eco-friendliness, and wide applications. However, proper system design and storage are required to overcome its limitations.

Advantages of Solar Energy

1. Renewable & Abundant

- Sunlight is available everywhere
- Unlimited energy source

2. Environment Friendly

- No air pollution
- No greenhouse gas emissions during operation

3. Low Operating Cost

- No fuel required
- Minimal maintenance cost

4. Suitable for Remote Areas

- Ideal for off-grid applications
- Useful in villages and hilly regions

5. Easy Installation

- Can be installed on:
  - Rooftops
  - Open lands
- Modular and scalable

6. Energy Independence

- Reduces electricity bills
- Promotes self-reliance

7. Versatile Applications

- Power generation
- Water heating
- Street lighting
- Agriculture (solar pumps)

8. Long Life

- Solar panels last 20–25 years

Disadvantages of Solar Energy

1. Intermittent Nature

Solar energy is available mainly during daytime.

- No sunlight during night.
- Cloudy and rainy conditions reduce output.
- Therefore, solar power generation is not continuous.

Example: Solar panels cannot generate electricity from sunlight at night.

2. Dependence on Weather

Solar power output depends strongly on solar radiation.

- Clouds reduce radiation.
- Rain reduces generation.
- Fog and dust can also reduce output.

Therefore, the power generated can vary throughout the day.

3. No Generation at Night

A PV system normally produces zero solar energy at night**.**

Therefore, if electricity is required at night, we need:

- Battery storage, or
- Grid connection, or
- Another generation source.

4. Energy Storage is Expensive

To use solar electricity when sunlight is unavailable, batteries may be required.

Batteries:

- Increase initial cost.
- Require maintenance/replacement.
- Have limited lifetime.
- Add weight and space requirements.

5. Large Land Requirement

Large-scale solar power plants require considerable land area.
This can create issues related to:

- Land availability
- Land-use competition
- Site preparation
- Ecological impacts

6. High Initial Cost

The initial investment can be significant because the system may require:

- PV modules
- Mounting structures
- Inverters
- Cables
- Protection equipment
- Batteries, if required
- Installation and civil works

Although operating costs are relatively low, the initial investment can be high.

7. Dust and Dirt Reduce Efficiency

Dust, dirt, bird droppings and other deposits can accumulate on the panel surface.
This reduces the sunlight reaching the solar cells and therefore can reduce electrical output.
Regular cleaning may be required, especially in dusty areas.

8. Lower Power Density

Solar radiation is spread over a large area.
Therefore, a relatively large collector area may be required to obtain substantial electrical power.
This is one reason large solar farms require significant land.

9. High Temperature Can Reduce PV Performance

PV module output is affected by temperature.
For many conventional crystalline-silicon PV modules:
As cell temperature increases, voltage generally decreases and the maximum power output tends to decrease.
Therefore, very hot operating conditions can reduce efficiency.

10. Environmental Issues Associated with Manufacturing and Disposal

Solar energy itself produces electricity without direct combustion emissions during operation.
However, environmental impacts can occur during:

- Raw-material extraction
- Manufacturing
- Transportation
- Installation
- End-of-life treatment and recycling

Proper recycling and responsible disposal of PV modules and batteries are therefore important.

International energy scenario of renewable sources

The leading countries for installed renewable energy in 2019 were China, the U.S., and Brazil. China was leading in renewable energy installations with a capacity of around 758.6 gigawatts. The U.S., in second place, had a capacity of around 264.5 gigawatts. Renewable energy is an important step in mitigating climate change and reducing the consequences caused by the phenomenon.

Country	Installed capacity in Gw
China	758.63
U.S.	264.5
Brazil	141.93
India	128.23
Germany	125.39
Canada	101
Japan	97.46
Italy	55.32
Russia	55.19
France	52.93

Table: 7. Installed capacity of renewable energy source across globe

Solar

1. Germany

At the end of May 2019, the cumulative solar power capacity of Germany reached 47.72 GW. The country has successfully met over 50% of the nation's daily energy demand from solar power.

2. China

Several centralized solar power projects of 2019 will get benefitted by China's government subsidies, like 1.7 billion Yuan (247.64 million dollars), involving the total installed capacity of 22.79 GW. According to reports from the National Energy Administration (NEA), 3,921 projects in 22 provinces and cities got approvals for these afore said subsidies.

3. Japan

Japan is still among the world's leaders in terms of total solar energy production, roughly around 55.5 GW in early 2019. The country could install nearly 155 GW by 2030 if things go according to their targets.

4. Italy

Likely to go some way to promoting renewable investment and helping Italy reach its 2030 targets are the seven competitive auctions to be held between 2019 and 2021, which include up to 4.8 GW in new PV and wind power plants, as well as 140 megawatts (MW) of hydro, biomass, and geothermal plants.

5. United States

According to the U.S Energy Information Administration, nearly 17% of electricity generation in the United States was hailed with the help of renewable power. All conditions remaining favourable, this perpetual growth might give rise to as much as four million solar installations by the year 2030. The U.S Department of Energy has also forecasted a 10% hike in their solar power generation by end of this year.

6. India

A country with one of the fastest-growing solar plants, India's solar installed capacity reached 28.18 GW in March 2019 and the country became the lowest cost producer of solar power in the world. The government had an initial target of 20 GW capacities for 2022, which was achieved four years ahead of schedule in 2018.

7. United Kingdom

Around 5% of Britain's total electricity generation was provided through solar by early 2019.

8. Australia

PV accounted for 5.2% of Australia's electrical energy production in early 2019, and as of March 2019, the country had over 12,035 MW of installed PV solar power, of which 4,068 MW were installed in the preceding 12 months. 59 solar PV projects with a combined installed capacity of 2,881 MW are also either under construction, constructed or due to start construction having reached financial closure.

9. France

Overall, the country's cumulative installed PV power surpassed an impressive 8.5 GW, with the newly installed PV capacity reaching 479 MW.

10. South Korea

According to the Ministry Of Trade, Industry & Energy, South Korea has successfully exceeded its Annual Deployment Target of 1.63 GW by installing 1.64 GW of Solar PV until July 2019. South Korea has plans to add 30 GW of PV by 2030, sundry of other steps have been undertaken to combat the country's overall former poor renewable performance, with 9% of that capacity to be developed in Saemangeum and 14GW of solar power energy to be installed before 2020.

Sustainable Design and development

Sustainable energy is a form of energy that meet our today's demand of energy without putting them in danger of getting expired or depleted and can be used over and over again. Sustainable energy should be widely encouraged as it do not cause any harm to the environment and is available widely free of cost. All renewable energy sources like solar, wind, geothermal, hydropower and ocean energy are sustainable as they are stable and available in plenty.

Sustainable energy sources

Fossil fuels are not considered as sustainable energy sources because they are limited, cause immense pollution by releasing harmful gases and are not available everywhere on earth. There are many forms of sustainable energy sources that can be incorporated by countries to stop the use of fossil fuels. Sustainable energy does not include any sources that are derived from fossil fuels or waste products. This energy is replenishable and helps to reduce greenhouse gas emissions and causes no damage to the environment. Hydropower is the most common form of alternative energy used around the world.

Need for Sustainable Energy

During ancient times, wood, timber and waste products were the only major energy sources. In short, biomass was the only way to get energy. When more technology was developed, fossil fuels like coal, oil and natural gas were discovered. Fossil fuels proved boom to the mankind as they were widely available and could be harnessed easily. When these fossil fuels were started using extensively by all the countries across the globe, they led to degradation of environment. Coal and oil are two of the major sources that produce large amount of carbon dioxide in the air. This led to increase in global warming. Also, few countries have hold on these valuable products which led to the rise in prices of these fuels. Now, with rising prices, increasing air pollution and risk of getting expired soon, forced scientists to look out for some alternative or renewable energy sources. Sustainable Energy came into the picture as it could meet our today's increasing demand of energy and also provide us with an option to make use of them in future also.

Sustainable Design

Sustainable design seeks to reduce negative impacts on the environment and the health and comfort of human beings, thereby improving performance of energy systems. Sustainable design principles include the ability to:

- optimize site potential;
- minimize non-renewable energy consumption;
- use environmentally preferable products;
- protect and conserve water;
- enhance indoor environmental quality; and
- optimize operational and maintenance practices.

Utilizing a sustainable design philosophy encourages decisions at each phase of the design process that will reduce negative impacts on the environment and the health of mankind, without compromising the bottom line. It is an integrated, holistic approach that encourages compromise and trade-offs. Such an integrated approach positively impacts all phases of an energy source life-cycle, including design, construction, operation and decommissioning.

- *25068 Solar domestic lighting systems installed in Tamil Nadu with assistance from Government.*
- *6095 Solar street lights installed in pubic places/streets mostly in village panchayats with Government assistance and active support and involvement of Rural Development Department.*

Development that meets present needs without compromising future generations**.**

Applications.

- **Residential Application:** Use of solar energy for homes has number of advantages. The solar energy is used in residential homes for heating the water with the help of solar heater. The photovoltaic cell installed on the roof of the house collects the solar energy and is used to warm the water. Solar energy can also be used to generate electricity. Batteries store energy captured in day time and supply power throughout the day. The use of solar appliances is one of the best ways to cut the expenditure on energy.
- **Industrial Application:** Sun's thermal energy is used in office, warehouse and industry to supply power. Solar energy is used to power radio and TV stations. It is also used to supply power to lighthouse and warning light for aircraft.
- **Remote Application:** Solar energy can be used for power generation in remotely situated places like schools, homes, clinics and buildings. Water pumps run on solar energy in remote areas. Large scale desalination plant also use power generated from solar energy instead of electricity.
- **Transportation:** Solar energy is also used for public transportation such as trolleys, buses and light-rails.
- **Pool heating:** Solar heating system can be used to heat up water in pool during cold seasons.

- **Solar Green Houses:** A green house is a structure covered with transparent material (glass or plastic) that acts as a solar collector and utilises solar radiant energy to grow plants. It has heating, cooling and ventilating devices for controlling the temperature inside the green house.
- **Solar Cooking:** The solar cooker requires neither fuel nor attention while cooking food and there is no pollution, no charring or overflowing of food and the most important advantage is that nutritional value of the cooked food is very high as the vita-mins and natural tastes of the food are not destroyed.
- **Solar furnace:** In a Solar furnace, high temperature is obtained by concentrating the solar radiations onto a specimen using a number of heliostats (turn-able mirrors) ar-ranged on a sloping surface.
- **Solar Drying of Agricultural and Animal Products:** Solar drying, especially of fruits improves fruit quality as the sugar concentra-tion increases on drying. Other agricultural products commonly solar-dried are potato-chips, berseem, grains of maize and paddy, ginger, peas, pepper, cashew-nuts, timber drying and tobacco curing. Spray drying of milk and fish drying are examples of solar dried animal products.
- **Solar-pumping:** In solar pumping, the power generated by solar-energy is utilized for pumping water for irrigation purposes.
- **Solar-distillation:** In arid semi and or coastal areas there is scarcity of potable water. The abundant sunlight in these areas can be used for converting saline water into portable distilled water by the method of solar distillation.

4.6.1 Fill Factor

Another term defining the overall behaviour of a solar cell is the fill factor (FF). It is a measure of squareness of the *I–V* characteristics of the solar cell and is defined as

$$\text{FF} = \text{Maximum output power}/(\text{open-circuit voltage} \times \text{short-circuit current})$$

It is the available power at the maximum power point (P_{MAX}) divided by the product of open-circuit voltage (V_{OC}) and short-circuit current (I_{SC}) as

$$\text{FF} = P_{MAX}/(V_{OC} \times I_{SC}) = (V_{MP} \times I_{MP})/(V_{OC} \times I_{SC}) \tag{4.17}$$

where V_{MP} and I_{MP} are the voltage and current at the maximum power point.

Equation (4.17) can be redefined as,

$$\text{FF} = (\eta \times S \times E)/(V_{OC} \times I_{SC}) \tag{4.18}$$

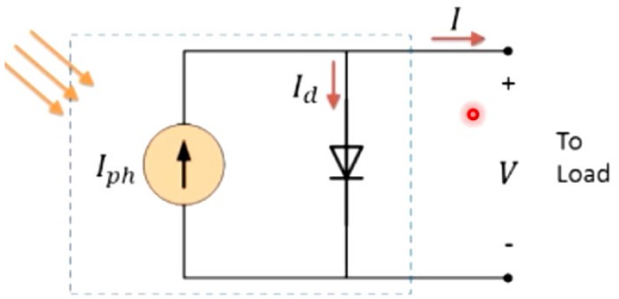
EQUIVALENT CIRCUIT AND THE CHARACTERISTICS OF A SOLAR CELL

First of all let us see the equivalent circuit of an ideal solar cell

Ideal solar cell means a solar cell that is having no losses that does not have any losses, a solar cell with no losses so if there are no losses that means there will not be resistances in the equivalent circuit. so ideal solar cell will not have any resistances.

so this is the equivalent circuit of an ideal solar cell

Equivalent Circuit of Ideal Solar Cell



Equivalent circuit of an ideal solar cell

The current flowing out of the solar cell is given as,

$$I = I_{ph} - I_0(e^{\frac{qV}{\alpha kT}} - 1)$$

I_{ph} - Photon current at a given irradiance ad given T

I_0 - Dark / Reverse saturation current of the diode

q – Charge of an electron $(1.602 \times 10^{-19} C)$

k – Boltzman's constant

T – Temperature in K

α – Diode ideality factor

It can be represented as a current source in parallel with a diode and this current source is denoted as Iph. which represents the photon generated current, the current is generated as a result of receiving solar radiation or photon and it is denoted as Iph and this current will be directly proportional to the irradiance, that is power in the solar cell so the photon current Iph is directly proportional to solar radiation the intensity of solar radiation or the irradiance that is power in the solar cell and the junction diode can be represented by a diode itself. The photon generated current flows and part of the current will flow through the diode and remaining obtained at the output so this I represent the output current or load current provided by a solar cell.

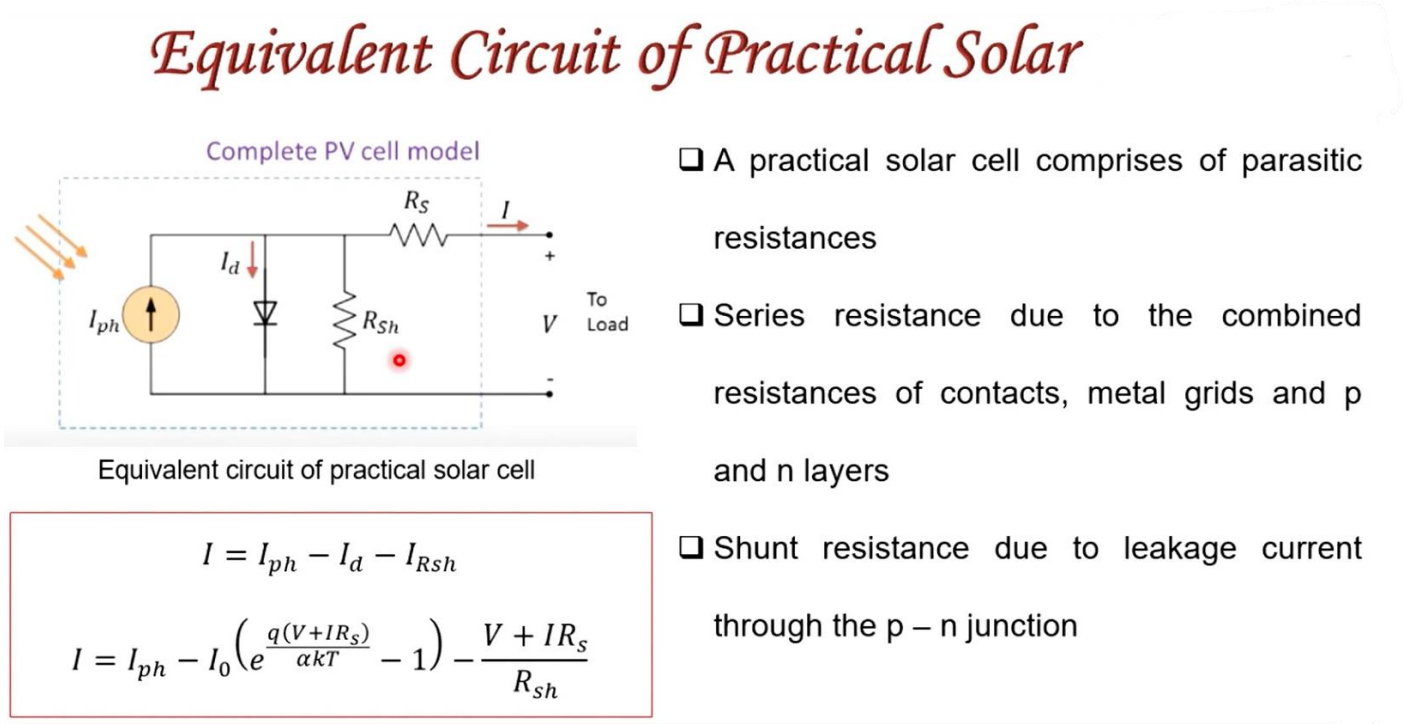
So we can write the equation for I,

$$I = I_{ph} - I_0(e^{\frac{qV}{\alpha kT}} - 1)$$

so here I_0 is the reverse saturation current of the diode and it can also be called as dark current. so when there is no solar radiation that is under dark condition, this diode Iph will be zero in that case no photon current will be there when there is no solar radiation so under that condition this diode will be reverse biased so the current flowing through the diode will be the reverse saturation current I_0 that is why I_0 is also known as dark current. so this is the equation of current obtained from a solar cell from an ideal solar cell.

Now we know that all solar cells are not ideal there will be losses so in order to account for these losses we have to add include resistances in the equivalent circuit.so now let us consider the equivalent circuit of a practical solar cell. so a practical solar cell comprises of parasitic resistances so definitely there will be resistances in the equivalent circuit of a practical solar cell.

So let us see how these resistances arises from where these resistances come into the equivalent circuit.So in the equivalent circuit we have series resistances as well as shunt resistances.



So we have seen the construction of a p-n junction solar cell, so we know that there are metallic contacts on the p layer and n layer , there were metal grids and also the p layer as well as the n layer have some resistances to the flow of current, so all these resistances are combined together and is represented as a series resistance R_S in the equivalent circuit, so a solar cell metal contacts and metal grids and then resistance of direction in addition to that appeal air semiconductor and layer semiconductor node resistance very physical resistance so all these resistance are combined together and this included as a series resistance in the equivalent circuit represented as $R_{S.}$

Now there will be a leakage current across the p-n junction across the junction or across the depletion layer there will be a leakage current and in order to represent this leakage current a shunt resistor is added because this current is flowing across the junction so we have to use a shunt resistor to represent this leakage current.so that is represented as $R_{Sh.}$ so we have to minimize the leakage current so this shunt resistor will have a higher value of resistance and the series resistance affects the drop of voltage,so it has to be minimized it will have a minimum value.

so in the equivalent circuit the series resistor will have a minimum value will have a small value and this shunt resistor will have a high value.so this is the equivalent circuit of a practical solar cell.

Characteristics of Solar Cell

Now we have to draw the characteristics of the solar cell,so a solar cell will be having two characteristics, the current versus voltage characteristics and the power versus voltage characteristics.

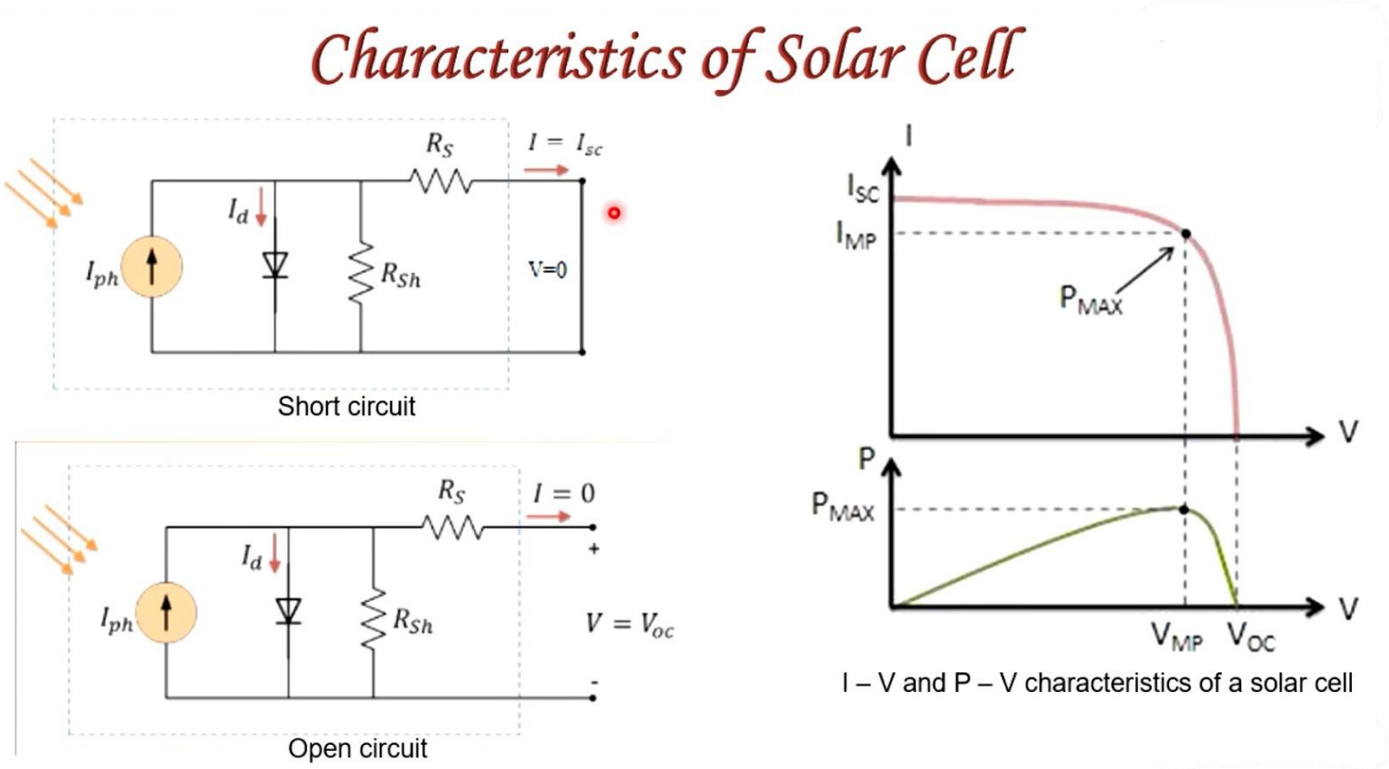
so IV and PV characteristics, we have to draw both characteristics are drawn against voltage.

current is drawn against voltage we have to plot power also against voltage.

Voltage has to be varied from a minimum value to the maximum value, from 0 to its maximum value we have to vary the voltage. In order to get zero voltage, we have to short circuit the output terminals, so voltage across a short circuit will always be zero, so zero voltage conditioning solar cylinder output in a short circuit, so voltage across this short circuit will be zero and through a short circuit maximum current will flow.

we know resistance of the short circuit will be zero,so current flowing through it will be maximum under short

circuit condition current flowing from the solar cell will be the maximum and this current can be represented as Isc.



so now what is the maximum value of voltage so we will get the maximum value of voltage when we open circuit the terminals of this solar cell, so corresponding to open circuit the voltage will be maximum.

which can be denoted as Voc the open circuit voltage and no current will flow through an open circuit, so current will be zero under this condition, so when voltage is maximum that is corresponding to Voc voltage becomes maximum the current will be zero. so we

will get the second point this particular point so these are two extreme points of this IV characteristics of a solar cell. And current Isc and point corresponding to zero current and voltage Voc and the remaining part of the graph resembles the forward biased characteristics of an ordinary p n junction diode so we have got the IV characteristics of a solar cell.

Now we have to draw the power versus voltage, calculate the power corresponding to any voltage.

power is equal to product of voltage and current and we have the VI characteristics with us.

so we have the value of current corresponding to all the voltages so from this graph calculate the power corresponding to different values of voltage at some random values of voltages

calculate the value of power as the product of voltage and corresponding current so now plot that values of power in another graph against the corresponding voltages and then while plotting a graph the shape will be like this so we can see that at a particular point the power is maximum and this point is known as maximum power point and the voltage corresponding to that maximum power point is marked as Vmp voltage corresponding to maximum power and the current corresponding to that maximum power is marked as Imp the current corresponding to maximum power. so this is the VI and characteristics of a solar cell first of all IV characteristics is plotted and from the characteristics we know the value of current at all voltages,so we can calculate the power corresponding to different voltages and we will plot that values of power against voltage in another graph so joining all those points we will get the power versus voltage curve and the point

at which the power is maximum is known as the maximum power point the corresponding voltage is marked as Vmp and corresponding current is marked as Imp. so this is all about the characteristics of a solar cell.

Solar energy collectors: Solar energy collectors are the device used for collecting the solar radiations and focus the solar radiations at particular location to transfer the heat energy into the solar ponds or fluid. Generally, two types of collectors are used first is non-concentrating or flat plate type solar collector which is used for low temperature cycle and second one is concentrating or focusing type solar collector which are used for medium and high temperature applications. Collectors make the solar energy more useful. Flat plate collectors are very simple, the collecting area is equal to absorbing area where as focusing type collector have several arrangements of mirrors and lenses for proper concentration of sun light. Due to this by using focus type collectors we can capture 100 times solar radiation as compared to flat plate collector keeping the area same. By using focusing type collector we can directly generate medium pressure steam.

Solar collectors are classified as

1. Non concentrating type
    - i) Flat-Plate Collectors
    - ii) Evacuated-Tube Collector
2. Concentrating type
    - i) Parabolic trough collector.
    - ii) Power tower receiver.
    - iii) Parabolic dish collector.
    - iv) Fresnel lens collector.

Flat-Plate Collectors

Flat-plate solar collectors are the most common ones. They consist of an absorber, a transparent cover and insulation. The main use of the technology is usually in residential buildings where the demand for hot water is big and affects bills. Commercial application of flat-plate collectors is usually seen in car washes, laundromats, military laundry facilities or restaurants.

The parts of a flat plate collectors are

- Black surface - absorbent of the incident solar energy
- Glazing cover - a transparent layer that transmits radiation to the absorber, but prevents radiative and convective heat loss from the surface
- Tubes containing heating fluid to transfer the heat from the collector
- Support structure to protect the components and hold them in place
- Insulation covering sides and bottom of the collector to reduce heat losses

Flat-plate solar collectors show a good price-performance ratio and also give a lot of mounting options.

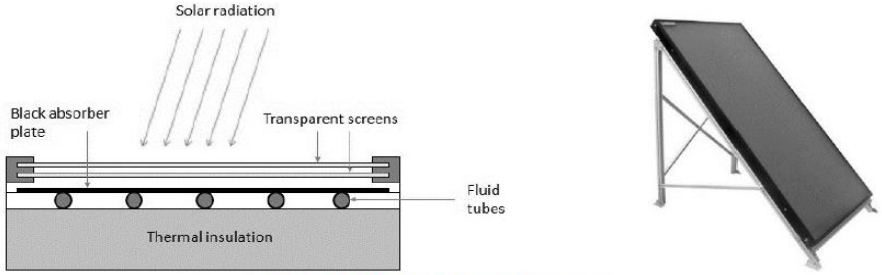


Fig. 9. Flat plate collectors

-Top 10 Solar Companies in India-
Adani Power, Tata Solar, Jinko Solar, Trina Solar, ACME Solar, Vikram Solar, Waaree Energies, EMMVEE, Goldi Solar, Canadian Solar

4.11. SOLAR WATER HEATING

Solar water heating is one of the most common applications of solar energy.

4.11.1. Natural Circulation Solar Water Heater

A natural circulation solar water is usually used in most "**domestic systems**".

A simple solar water heater with natural circulation is shown in Fig. 4.5.

Construction. It consists of a *flat-plate solar collector*, normally single glazed, and insulated *storage tank* kept at a height. It is installed on a roof with the collector facing the sun and connected to cold water mains. The collector comprises *copper tubes welded* to a *copper sheet* (both coated with a highly absorbing *black coating*) with a *toughened glass sheet* on top and insulating material on the rear.

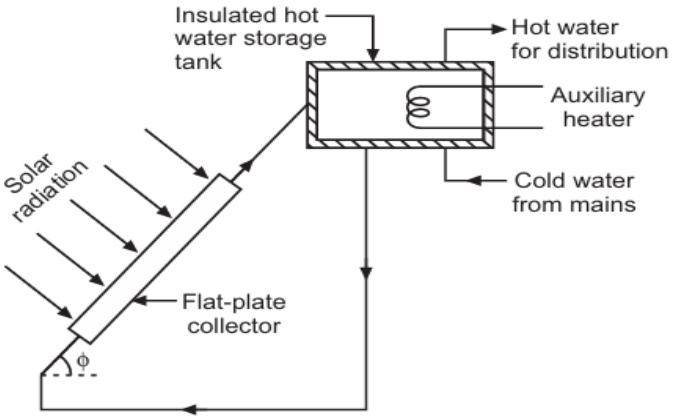


Fig. 4.5. Natural circulation solar water heater.

Working. The collector collects the sun radiation due to which water flowing through the tubes absorbs solar heat and is stored in a tank. The water circulation is entirely *based on the density difference between the solar-heated water in the collector and the cold water in the storage tank*. Hot water for use is taken out from the top of the tank.

To provide heat during long, cloudy periods, an *electrical immersion heater (i.e. 'auxiliary heater')* can be used as a back-up for the solar system. A non-freezing fluid may be used in the collector circuit.

- *Most domestic systems are of capacity ranging from 100 to 150 litres of hot water per day.*

Evacuated-Tube Collector

This is a type of a vacuum collector. Its absorber strip is placed in an evacuated and pressure proof glass tube. The heat transfer fluid flows directly the absorber into a U-tube or in a tube-in-tube system. The heat pipe collector integrates a special fluid, which evaporates even at low temperatures, thus the steam rises in the individual heat pipes and warms up the fluid in the main pipe, generating heat. Thermodynamic panels

are also based on such a refrigerant fluid but are exploiting the heat in the ambient air, and, therefore, are only suitable for hot water.

The technology is very reliable as it has an estimated lifespan of 25 years. The vacuum that surrounds the outside of the tubes greatly reduces the risk of heat loss, therefore efficiency is greater than it is with flat-plate collectors.

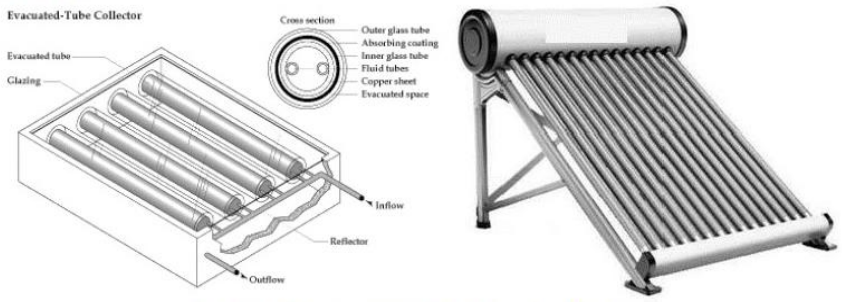


Fig. 10. Evacuated-Tube Collector

Step-by-Step Working

1. Solar radiation passes through the glass cover
2. Absorber plate absorbs radiation and gets heated
3. Heat is transferred to the fluid (water/air) in tubes
4. Heated fluid flows to storage tank or application
5. Insulation minimizes heat losses

This works based on the greenhouse effect**:**

- Shortwave radiation enters
- Longwave heat radiation is trapped

Key Features

- Operating temperature: 30°C to 80°C
- No tracking required
- Simple construction

Applications of Non-Concentrating Collectors

1. Solar Water Heating

- Domestic and industrial use

2. Space Heating

- Buildings, greenhouses

3. Air Heating

- Crop drying, ventilation systems

4. Industrial Heating

- Low-temperature processes

5. Desalination

- Solar stills

6. Swimming Pool Heating

- Maintaining water temperature

Advantages

Simple and Low Cost
Easy Installation and Maintenance
No Sun Tracking Required
Reliable and Durable

Limitations

Lower Efficiency Compared to Concentrating Collectors
Limited to Low Temperatures
Larger Area Required
Heat Losses Due to Environment

Non-concentrating solar collectors, especially flat plate collectors, are widely used for low-temperature applications due to their simplicity, reliability, and cost-effectiveness. They operate on the greenhouse effect and are ideal for domestic and small-scale industrial heating needs.

Concentrating solar collectors

Parabolic Trough Collector

- It is a line focusing type collector. In this type of collector, the solar radiations falling on the area of the parabolic reflector are concentrated at the focus of the parabola.
- When the reflector is manufactured in the form of a trough with the parabolic cross-section, the solar radiations gets focused along a line. An absorber pipe is placed along this line and a working fluid (usually synthetic oil or water) flows through it.
- When the focused solar radiations fall on the absorber pipe, it heats the fluid to a high temperature. Then the heat absorbed by the working fluid is transferred to water for producing steam.
- The focus of solar radiations changes with the change in sun's elevation. In order to focus the solar radiations on the absorber pipe, either the trough or the collector pipe is rotated continuously about the axis of the absorber pipe.

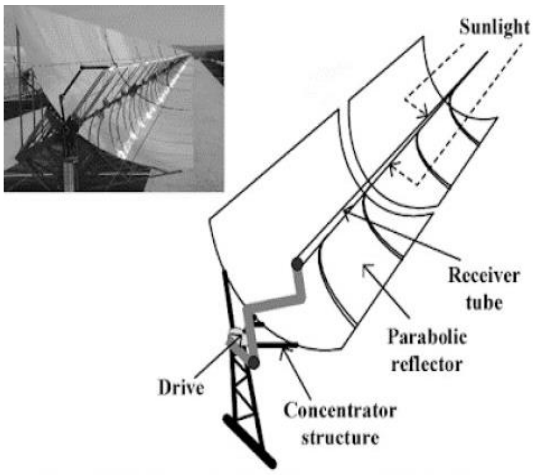


Fig.11. Parabolic Trough Collector

Power Tower Receiver

In this collector, the receiver is located at the top of the tower. It has a large number of independently-moving flat mirrors (heliostats) spread over a large area of ground to focus the reflected solar radiations on the receiver. The heliostats are installed all around the central tower. Each heliostat is rotated into two directions so as to track the sun. The solar radiations reflected from heliostats are absorbed by the receiver mounted on a tower of about 500 m height. The tower supports a bundle of vertical tubes containing the working fluid. The working fluid in the absorber receiver is converted into the high-temperature steam of about 600°C – 700°C. This steam is supplied to a conventional steam power plant coupled to an electric generator to generate electric power.

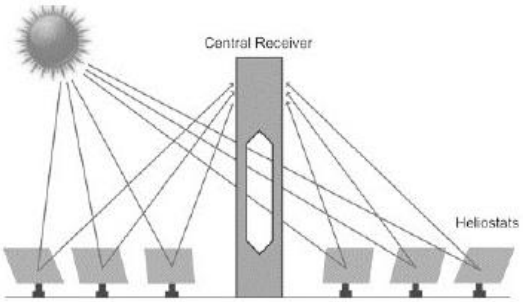


Fig.12. Power tower receiver

Parabolic Dish Collector

- In these collectors, the receiver is placed at the focal point of the concentrator. The solar beam radiations are focused at a point where the receiver (absorber) is placed. The solar radiations are collected in the receiver.
- A small volume of fluid is heated in the receiver to a high temperature. This heat is used to run a prime mover coupled with a generator.
- A typical parabolic dish collector has a dish of 6 m diameter. This collector requires two-axis tracking. It can yield temperatures up to 3000° C.
- Due to the limitations of size and the small quantity of fluid, dish type solar collectors are suitable for only small power generation (up to few kW).

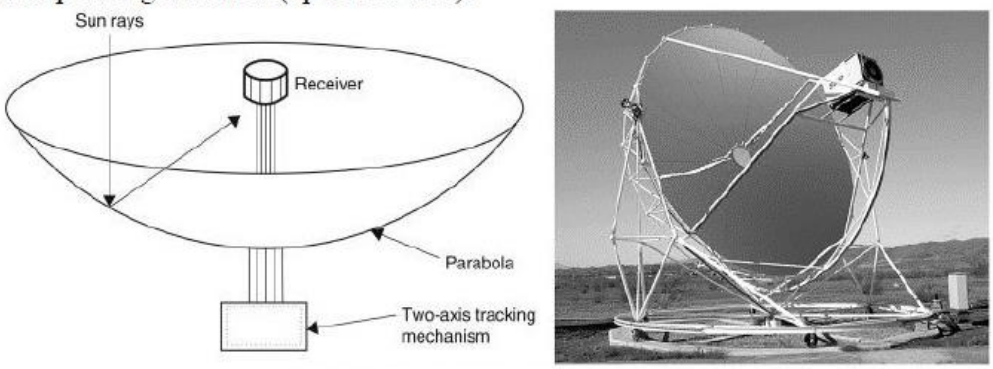


Fig.13. Parabolic Dish Collector

Applications of Concentrating Collectors

1. Solar Thermal Power Generation
- Used in steam turbines (CSP plants)

2. Industrial Process Heat
- Chemical, textile, and food industries

3. Steam Generation
- Boilers and thermal plants

4. Solar Cooling Systems
- Absorption refrigeration

5. Desalination
- Large-scale water purification

6. High-Temperature Research
- Material testing, furnaces

Advantages

High Temperature Output
Higher Efficiency
Suitable for Power Generation

Limitations

High Initial Cost
Requires Tracking System
Needs Clear Sky (Direct Radiation only)
Complex Design and Maintenance

Solar collectors are broadly classified into non-concentrating and concentrating types**.** Concentrating collectors use mirrors or lenses to focus sunlight**,** enabling high-temperature applications like power generation and industrial heating, making them essential in large scale solar energy systems.

Working of solar power plant

The working is very simple almost similar to any thermal power plant. Solar power plant also works on Rankine cycle and Brayton cycle as per the requirements. With the help of construction, we can easily predict the cycle of operation and working. Working fluid gains latent heat of vaporisation from the direct solar radiations or by means of solar ponds in the evaporator boiling and converts it into vapour form. After that it runs the turbine which is connected to the generator. Then the turbine working fluid goes into the condenser and loses heat and again sends back to the evaporator boiler with the help of pump. This whole cycle repeats continuously until the sun remains in the sky and radiation falls on the earth surface.

Advantages:
- Solar power plants work on solar energy which is available in abundant on the earth surface at most of the places.
- Solar power plants produce negligible pollution as compared to thermal power plant.
- The energy produced is renewable energy with negligible cost.
- Quantity of water used in solar power plants is very less as compare to other power plants.

Disadvantages:
- The major drawback is the availability of the sun. Sun's radiation of desired intensity is not available whole day.
- For collecting the sun radiations at useful rate large area is required.
- Initial setup cost of Solar plant is quite high.

It would cost anywhere between Rs 70,000 to Rs 1,20,000 per kW depending on the panels and inverter

Solar Thermal Power Plant

Working Principle

A solar thermal power plant converts solar energy into electrical energy using heat as an intermediate form.

Step-by-Step Operation:

1. Solar radiation is collected using mirrors/collectors
2. Sunlight is concentrated to produce high-temperature heat

3. Heat is used to generate steam in a boiler/heat exchanger
4. Steam drives a turbine connected to a generator
5. Steam is condensed and recycled (Rankine cycle)

It works on the principle of converting **solar → thermal → mechanical → electrical energy**

Advantages of Solar Thermal Power Plants

Renewable and Clean Energy

- No fuel cost and low emissions

Suitable for Large-Scale Power Generation

- Can generate electricity in MW range

Thermal Energy Storage Possible

- Enables power generation even at night

Reduced Dependence on Fossil Fuels

Disadvantages

High Initial Cost

- Expensive setup and infrastructure

Requires Large Land Area

Depends on Weather Conditions

- Efficiency drops during cloudy days

Needs Direct Sunlight

- Not effective in low solar radiation regions

Water Requirement

- Needed for steam generation and cooling

Central Receiver Power Plants

Unlike linear concentrating systems (troughs), which reflect light onto a focal line, the central receiver systems send concentrated light onto a remote central receiver. A typical example of such a system is a solar power tower system, which consists of multiple tracking mirrors (heliostats) positioned in the field around a main external receiver installed on a tower. Such systems are capable of reaching of much higher levels of concentration than linear systems. Concentrated radiation is further used as heat to produce steam and convert it to electricity (like in a regular power plant), or the generated thermal energy can be stored in a molten salt storage.

Central receiver systems are typically large-scale plants that are usually built to power a steam cycle. The central position of the receiver offers a universal advantage to collect all energy at one location and save on transport networks. Central receiver systems use a field of distributed, circular array of mirrors that is, heliostats which individually track the sun and focus the sunlight on the top of a tower. By concentrating the sunlight 600 – 1000 times, they achieve temperatures from 800°C to well over 1000°C. Solar energy is absorbed by a working fluid and then used to generate steam to power a conventional turbine. The high temperatures available in solar towers can be used not only to drive steam cycles, but also for gas turbines and combined cycle systems. Such systems can achieve up to 35% peak and 25% annual solar electric efficiency when coupled to a combined cycle power plant.

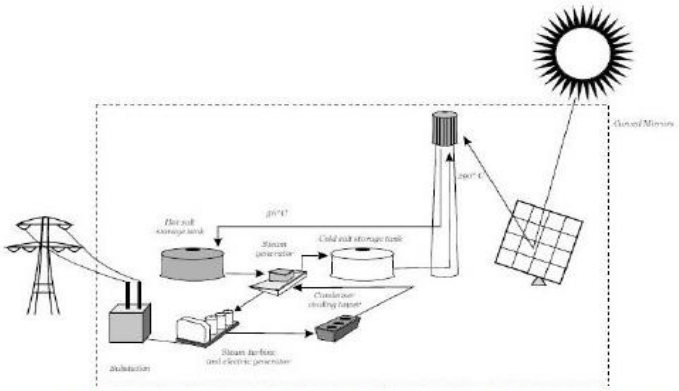


Fig.15. Operation of Central Receiver Power Plants

Major sub-components of central receiver system

- **Heliostat field:** The heliostat field comprises a large heliostat, structure, and control/tracking. The heliostat typically utilizes a mirror, which can be oriented throughout the day to redirect sunlight along a fixed axis toward a stationary target or receiver. The reflecting element of a heliostat is typically a thin, back (second) surface, low-iron glass mirror. This heliostat is composed of several mirror module panels rather than a single large mirror. The thin glass mirrors are supported by a substrate backing to form a slightly concave mirror surface. Individual panels on the heliostat are also canted toward a point on the receiver. The heliostat focal length is approximately equal to the distance from the receiver to the farthest heliostat. Subsequent "tuning" of the closer mirrors is possible.

- **Storage:** Central tower based systems typically use Molten salt, hot concrete storage, phase change materials, saturated steam or pressurized air as storage media.
  - Nitrate salt mixtures can be used as both a heat transfer fluid and a storage medium at temperatures of up to 565°C. However, most mixtures currently being considered freeze at temperatures around 140 to 220°C and thus must be heated when the system is shutdown. They have a good storage potential because of their high volumetric heat capacity.
  - Liquid sodium can also be used as both a heat transfer fluid and storage medium, with a maximum operating temperature of 600°C. Because sodium is liquid at this temperature, its vapour pressure is low. However, it solidifies at 98°C thereby requiring heating on shutdown. The cost of sodium-based systems is higher than the nitrate salt systems.
  - For high-temperature applications such as Brayton cycles, it is proposed to use air or helium as the heat transfer fluid. Operating temperatures of around 850°C at 12 atm pressure are being proposed. Although the cost of these gases would be low, they cannot be used for storage and require very large diameter piping to transport them through the system.

- **Receivers/absorber and power block:** This includes the receivers, absorbers including heat collection elements, and Power Block. The receiver, placed at the top of a tower, is located at a point where reflected energy from the heliostats can be intercepted most efficiently. The receiver absorbs the energy being reflected from the heliostat field and transfers it into a heat transfer fluid. There are two basic types of receivers: external and cavity.
  - **External Receivers.** These normally consist of panels of many small (20-56 mm) vertical tubes welded side by side to approximate a cylinder. The bottoms and tops of the vertical tubes are connected to headers that supply heat transfer fluid to the bottom of each tube and collect the heated fluid from the top of the tubes.
  - **Cavity Receivers.** In an attempt to reduce heat loss from the receiver, some designs propose to place the flux absorbing surface inside of an insulated cavity, thereby reducing the convective heat losses from the absorber. The flux from the heliostat field is reflected through an aperture onto absorbing surfaces forming the walls of the cavity. Typical designs have an aperture area of about one-third to one-half of the internal absorbing surface area. Cavity receivers are limited to an acceptance angle of 60 to 120 degrees. Therefore, either multiple cavities are placed adjacent to each other, or the heliostat field is limited to the view of the cavity aperture.

Working Principle

Step-by-Step Process:

1. Sunlight falls on heliostat mirrors
2. Mirrors track the sun and reflect rays to the central receiver
3. Receiver absorbs intense heat (up to 1000°C)
4. Heat is transferred to molten salt/HTF
5. Steam is produced in a heat exchanger
6. Steam drives the turbine-generator

7. Exhaust steam is condensed and reused

Key Features

- Very high temperature operation
- High efficiency compared to other CSP systems
- Can include thermal energy storage

Applications

Large-Scale Electricity Generation
Industrial High-Temperature Heat
Energy Storage-Based Power Supply

A solar thermal power plant efficiently converts solar energy into electricity using thermal processes. The central receiver system is one of the most advanced CSP technologies, offering high efficiency and large-scale power generation**,** though it comes with higher cost and complexity.

Solar Energy Storage Systems and Applications

Solar Energy Storage is the process of collecting solar energy during sunny hours and storing it for use when sunlight is unavailable.

Suppose
Morning - Plenty of sunlight
Night - No sunlight

Without storage
No electricity at night
Solar pumps stop
Solar street lights will not work

Types of Solar Energy Storage

Thermal Storage

Electrical Storage

Mechanical Storage

Chemical Storage

1. Thermal Energy Storage (Heat Storage)
Heat obtained from the Sun is stored for later use.

Examples
•Water
•Molten Salt
•Rocks
•Sand

Applications
Solar Water Heater
Solar Cooker
Solar Dryer

Advantages
Simple
Low Cost
Easy to Maintain

2.Electrical Energy Storage
Solar electricity is stored in batteries.

•Solar Home Systems
•Solar Street Lights
•UPS Systems

Batteries Used
•Lead Acid Battery
•Lithium-ion Battery
•Nickel-Cadmium Battery

Advantages
Reliable
Easy to use
Most common method

3. Mechanical Energy Storage

Solar energy is first converted into mechanical energy and then stored.

Examples:

•Pumped Hydro Storage
•Flywheel
•Compressed Air Energy Storage

4. Chemical Energy Storage

Solar electricity is used to produce hydrogen.

This is called Hydrogen Energy Storage.

Storage Method	What is Stored?	Example
Thermal	Heat	Hot Water
Electrical	Electricity	Battery
Mechanical	Mechanical Energy	Pumped Hydro
Chemical	Hydrogen	Fuel Cell

Applications of Solar Energy Storage

1.Solar Street Lights
2. Solar Home System
3. Solar Water Heating
4. Solar Irrigation Pump
5. Solar Cooker
6. Solar Dryer

Advantages of Solar Energy Storage

Continuous power supply
Energy available at night
Reduces electricity bills
Increases reliability
Supports renewable energy

Limitations

High installation cost
Batteries require replacement
Storage losses occur
Large systems need more space

Terrestrial Solar Radiation & Extra-terrestrial Solar Radiation

Some sunlight is absorbed.
Some is reflected.
Some is scattered.
Only part of it reaches the Earth's surface.
This leads to two important terms,

- Extra-terrestrial Solar Radiation
- Terrestrial Solar Radiation

Terrestrial Solar Radiation

- Terrestrial Solar Radiation is the solar radiation that reaches the Earth's surface after passing through the atmosphere.

When sunlight enters the atmosphere,
it undergoes

- Absorption
- Scattering
- Reflection

Only part of the original radiation reaches Earth.

Atmospheric gases absorb energy.
Examples:
Ozone-Absorbs UV Rays
Water Vapour-Absorbs Infrared
CO_2-Absorbs Long-wave Radiation
Dust-Scatters Sunlight
Clouds-Reflect Sunlight

Process	Energy Loss
Absorption	10–30%
Reflection	2–11%
Scattering	5–26%

So, Earth receives much less energy than exists outside the atmosphere.

Types of Terrestrial Radiation

1. Beam (Direct) Radiation
Comes directly from the Sun.
Travels in straight lines.
Can be concentrated.
Example:
Solar concentrators

2. Diffuse Radiation
Produced due to scattering.
Comes from all directions.
Cannot normally be concentrated.

3. Global Radiation
Global Radiation
=
Beam Radiation
+
Diffuse Radiation

Extra-terrestrial Solar Radiation

Extra-terrestrial Solar Radiation is the solar radiation received outside the Earth's atmosphere.
It is the original solar energy coming directly from the Sun.

Where is it Measured?
Outside the atmosphere
At the top of Earth's atmosphere
Approximately at 1 Astronomical Unit (1 AU)

Characteristics
Maximum solar intensity
No atmospheric losses
No scattering
No absorption
No reflection

Why is it Called "Extra-Terrestrial"?
Extra = Outside
Terrestrial = Earth
So,
Outside Earth
=
Extra-terrestrial

Extraterrestrial Radiation	Terrestrial Radiation
Outside atmosphere	At Earth's surface
Maximum intensity	Lower intensity
No atmospheric losses	Atmospheric losses present
Solar Constant	Actual available solar energy

AM0	AM1 and above

Applications

Extra-terrestrial Radiation

- Satellite calculations
- Space research
- Solar constant calculations

Terrestrial Radiation

Used in

- Solar Panels
- Solar Water Heater
- Solar Thermal Plants
- Solar Pumping
- Solar Dryers

Solar pond: The solar energy coming from the sun is firstly absorbed by the solar pond. Solar pond is a reservoir of water where sun's rays directly focus by using solar energy collectors. After that, solar energy increases the internal energy and raises the temperature of solar pond. The fluid in the solar pond may be directly used as the working fluid if its temperature reaches up to evaporating temperature or other way is to use a secondary fluid known as working fluid. The fluid from the solar pond goes into cyclic process. The detailed operation of solar pond is as follows.

Solar pond, also called solar 'salt pond', is an artificially designed pond, filled with salty water, maintaining a definite concentration gradient. It combines solar energy radiation and sensible heat storage, and as such, it is utilised for collecting and storing solar energy. A solar pond reduces the convective and evaporative heat losses by reversing the temperature gradient with the help of non-uniform vertical concentration of salts.

The vertical configuration of "salt gradient solar pond" normally consists of the following three zones:

1. "Surface (homogeneous) convective zone (SCZ)"- It is adjacent to the surface and serves as a buffer zone between environmental fluctuations at the surface and conductive heat transport from the layer below. It is about 10 to 20 cm thick with a low uniform concentration at nearly the ambient air temperature.
2. "Lower connective zone (LCZ)"- It is at the bottom of the pond and this is the layer with highest salt concentration, where high temperatures are built up.
3. "Concentration/Intermediate gradient zone (CGZ)"- This zone keeps the two convective zones (SCZ and LCZ) apart and gives the solar pond its unique thermal performance. It provides excellent insulation for the storage layer, while transmitting the solar radiation. To maintain a solar pond in this non-equilibrium stationary state it is necessary to replace the amount of salt that is transported by molecular diffusion from the LCZ to SCZ. This means that salt must be added to the LCZ, and fresh water to the SCG whilst brine is removed. The brine can be recycled, divided into water and salt (by solar distillation) and returned to the pond.

The major heat loss occurs from the surface of the solar pond. This heat loss can be prevented by spreading a plastic grid over the pond's surface to prevent disturbance by the wind. Disturbed water tends to lose heat transfer faster than when calm.

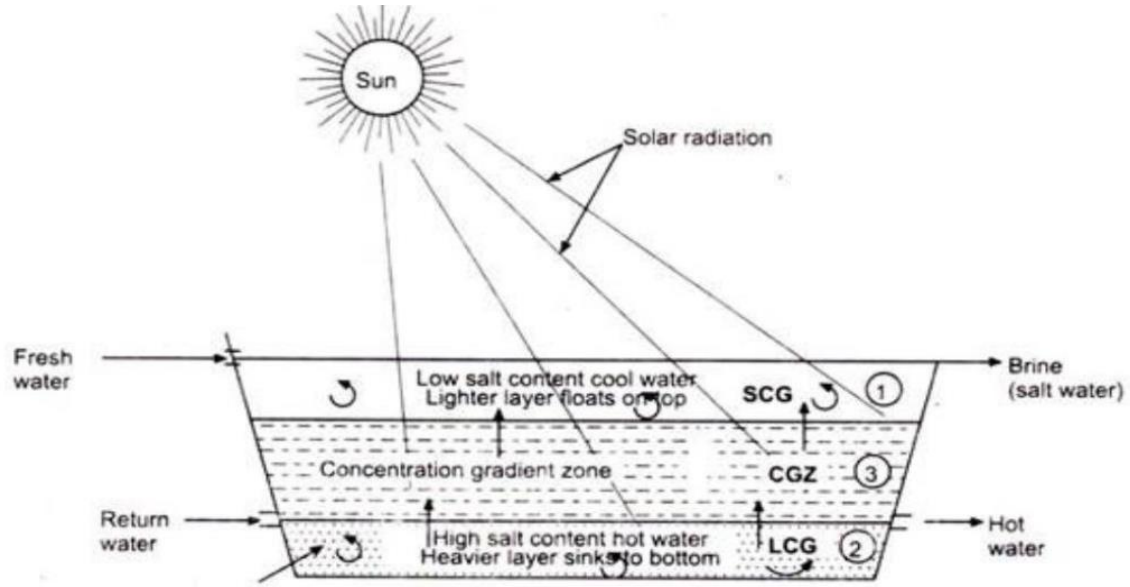


Fig.8. Principle of Solar Pond

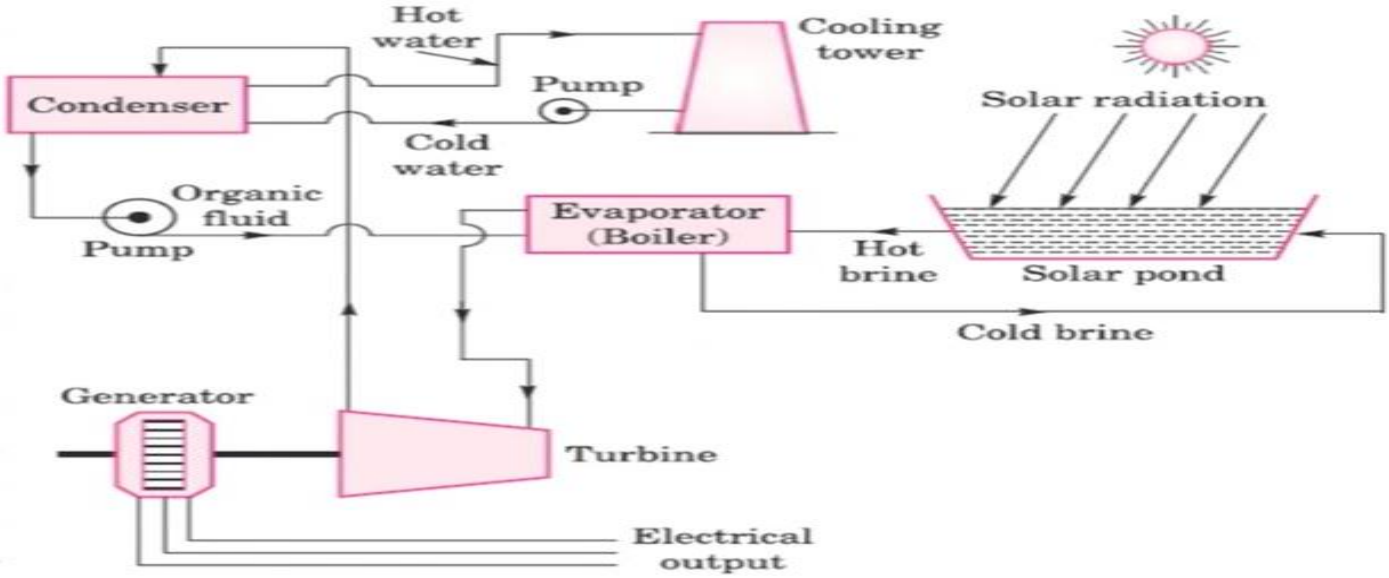


Fig. 4.6. Solar pond electric power plant with, cooling tower.

Principle of Solar Pond

Due to the excessively high salt concentration of the LCZ, a plastic liner or impermeable soil must be used to prevent infiltration into the nearby ground water or soil. The liner is a factor that increases the cost of a solar pond. A site where the soil is naturally impermeable, such as the base of a natural pond or lake, or can be made impermeable by compaction or other means, will allow considerably lower power costs.

The optical transmission properties and related collection efficiency vary greatly and depend on the following factors:

i. Salt concentration.
ii. The quantity of suspended dust or other particles.
iii. Surface impurities like leaves or debris, biological material like bacteria and algae.
iv. The type of salt.

It becomes obvious that much higher efficiencies and storage can be achieved through the utilization of refined or pure salt whenever possible, as this maximizes optical transmission.

The solar pond is an effective collector of diffuse, as well as direct radiation, and will gather useful heat even on cloudy or overcast days. Under ideal conditions, the pond's absorption efficiency can reach 50% of incoming solar radiation, although actual efficiencies average about 20% due to heat losses.

Temperature Distribution

- **Top layer (UCZ):** ~25–30°C (near ambient)
- **Middle layer (NCZ):** Temperature increases gradually
- **Bottom layer (LCZ):** ~70–90°C

Temperature increases with depth due to:

- Solar absorption at the bottom
- No convection to dissipate heat

Advantages of Solar Ponds

Simple and Low Cost

- No complex equipment required

Energy Storage Capability

- Stores heat for **day and night use**

Large-Scale Heat Collection

- Suitable for industrial applications

Renewable and Eco-Friendly

- Uses solar energy

Applications of Solar Ponds:

1. Power generation.
2. Space heating and cooling.
3. Crop drying.
4. Desalination.
5. Process heat.

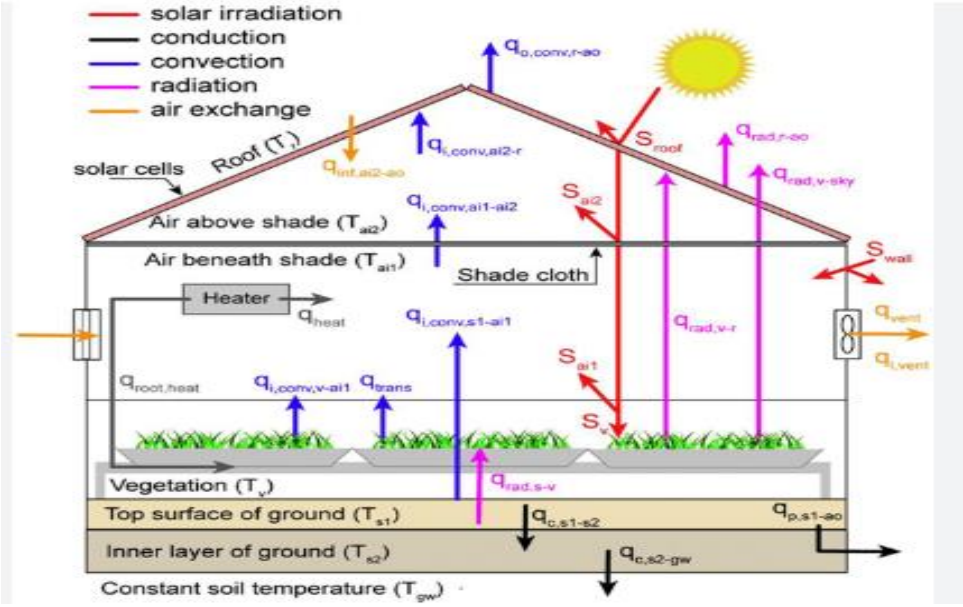
Limitations of Solar Ponds:

1. Sunny climate is required
2. Need for large land area.
3. Availability of salt.
4. Availability of water.

> *Solar energy is a completely free source of energy and it is found in abundance. Though the sun is 90 million miles from the earth, it takes less than 10 minutes for light to travel from that much of distance.*

A solar pond is an innovative and economical method for collecting and storing solar thermal energy using a salt gradient to suppress convection**.** While it is ideal for large-scale, low-temperature applications, its limitations restrict its use in compact or high-efficiency systems.

Solar Green house



This diagram represents a thermal energy balance and heat transfer model of an engineered, solar-powered greenhouse. It illustrates how energy moves via radiation, conduction, convection, and air exchange between the structural components, plants, and the environment.

Construction:

Solar Cell Roof (T_r): A sloped roof integrated with solar cells to generate electricity while allowing selective solar irradiation (S_{roof}) to enter.

Divided Air Zones: A horizontal **shade cloth** splits the internal volume into two distinct microclimates: "Air above shade" (T_{ai2}) and "Air beneath shade" (T_{ai1}).

Active HVAC System:

- An automated **ventilation fan** system on the sidewalls regulates air exchange (q_{vent}).
- An auxiliary **heater** provides both space heating (q_{heat}) and targeted root-zone heating ($q_{\text{root,heat}}$).

Vegetation Level (T_v): Elevated plant benches or trays positioned above the ground layer.

Stratified Ground Layer: A multi-tiered flooring system consisting of the top surface of the ground (T_{s1}), an inner layer of ground (T_{s2}), resting over deep soil with a constant temperature (T_{gw}).

Operations:

The greenhouse regulates its internal climate through five distinct energy pathways:

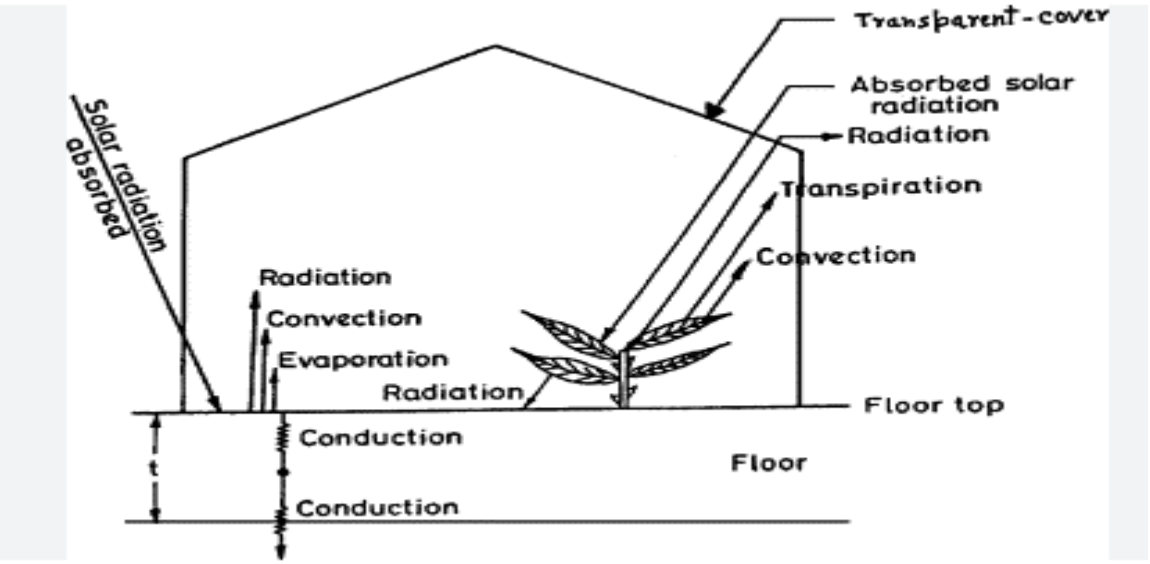
Solar Irradiation (Red Arrows): Sunlight penetrates the roof (S_{roof}) and walls (S_{wall}), distributing solar energy directly to the plants (S_v), the internal air zones, and the ground surface.

Convection (Blue Arrows): Heat naturally transfers through air currents between the plants, soil surface, the shade cloth boundaries ($q_{i,\text{conv}}$), and escapes through the outer roof ($q_{o,\text{conv}}$).

Radiation (Pink Arrows): Long-wave thermal radiation is exchanged internally between the ground and vegetation ($q_{\text{rad},s-v}$), and lost from the roof and interior out into the open sky ($q_{\text{rad},v-\text{sky}}$).

Conduction (Black Arrows): Heat moves physically downwards through the earth layers ($q_{c,s1-s2}$) to store or dissipate energy into the deep ground.

Air Exchange (Orange Arrows): Mechanical fans pull in fresh air and push out stale, overheated air (q_{vent}) to manage humidity and temperature.



Advantages of Solar Greenhouse

1. **Uses renewable solar energy**
   It uses sunlight, which is a clean and renewable source of energy.
2. **Reduces heating cost**
   Solar heat can reduce the need for conventional heating systems, especially during cold periods.
3. **Maintains suitable temperature**
   The greenhouse helps maintain a favourable temperature for plant growth.
4. **Extends the growing season**
   Crops can be grown for a longer period, including during colder seasons.
5. **Improves crop productivity**
   Controlled environmental conditions can improve plant growth and productivity.
6. **Protects crops from adverse weather**
   Plants are protected from conditions such as cold winds, frost, heavy rain and some pests.
7. **Better water utilization**
   Controlled greenhouse conditions can reduce unnecessary water loss and improve water use efficiency.
8. **Suitable for off-grid locations**
   Solar-based systems can be useful in remote areas where conventional energy supply is limited.
9. **Reduces fossil-fuel consumption**
   Using solar energy reduces dependence on conventional fuels for heating and other greenhouse requirements.
10. **Environmentally friendly**
    Solar energy produces no direct combustion emissions during operation.